LIGHTYEAR

MS 105 Reproducible compiler builds and MCP recovery

Recorded October 14 2026. Delivery: PR #338 and the recovery repair in PR #337.

Two infrastructure repairs make verification more dependable. The pinned compiler image can install its verified packages, and a status check can reconnect to a running workflow without reading an unfinished journal database. These repairs improve execution reliability while retaining the existing evidence checks.

Compiler image repair

APT's restricted package-download user could not read staged package files. The installer now gives the staging directory and verified packages the access needed for installation and explicitly checks readability. Snapshot access uses HTTPS, bounded retries and timeouts, and fails when the package index cannot be updated.

Compiler versions, package hashes, snapshot date and base-image pins remain unchanged. The repair therefore addresses access and transport failures without substituting a different compiler. PR #338 passed all 29 checks on commit `679647d0caf425b3203a8874f304d11ac1c76d2a` and merged as `544247928a871bca4ecb9971185584569a51cdad`.

MCP disconnect recovery

The workflow journal had a startup race: SQLite created a visible file before its events table existed. A concurrent status reader could open that incomplete file and reject the run. New journals are now initialized in a temporary file and published atomically after the database is committed and closed, under the existing writer lease.

Existing journals are preserved. A failed publication removes the temporary file and releases the lease. Missing schema, corrupt databases and invalid evidence still fail verification. The repair does not suppress errors or turn an invalid journal into an empty successful run.

Validation

Commit `4c66d2ed39f7d9d99cd3086bc5696a3acbf56328` passed the local-agent CI suite on Windows 2025 and Ubuntu 24.04. Each platform passed four deterministic journal tests plus all 26 existing workflow and MCP tests. The previously failing test disconnected the real MCP server, observed the independent worker and reconnected to verify its retained result.

The deterministic test pauses creation before the schema exists and verifies that readers cannot see the temporary database. The original Windows error concealed its underlying exception, so the measured claim is that this race is covered and the previously failing recovery test now passes.

Supporting evidence

- Compiler build and twin reproducibility
- Windows and Linux recovery tests
- Atomic journal implementation
- Journal regression tests

Passing infrastructure checks does not approve a customer release or replace workload-specific equivalence evidence.